

2023-2024 Fall
MAT124 Final
11/01/2024

1. a. Find the critical points of the function

$$f(x, y) = e^y (y^2 - x^2)$$

and classify them.

- b. Find the maximum and minimum values of the function

$$f(x, y, z) = xyz^2$$

by using Lagrange multipliers on the region $D : x + y + z = 1, x \geq 0, y \geq 0, z \geq 0$.

2. a. Sketch the domain of integration, and rewrite the integral by changing the order of integration.

$$\int_0^1 \int_{\sqrt{1-y}}^1 f(x, y) dx dy + \int_1^3 \int_{\frac{y-1}{2}}^1 f(x, y) dx dy$$

- b. Evaluate the integral

$$\iint_R \sqrt{x^2 + y^2} dA$$

where R is the region bounded by the line $x = 1$ on the left and by the circle $x^2 + y^2 = 2$ on the right.

3. Convert

$$\int_0^{2\pi} \int_0^{\sqrt{2}} \int_r^{\sqrt{4-r^2}} 3 dz r dr d\theta, \quad r \geq 0$$

- a. to rectangular coordinates with the order of integration $dz dx dy$.
b. to spherical coordinates.
4. a. Is $\mathbf{F}(x, y) = 2xy \cos(x^2y) \mathbf{i} - x^2 \cos(x^2y) \mathbf{j}$ conservative? Why?
b. Show that

$$\begin{aligned} \mathbf{F}(x, y, z) = & (y^2 \cos(xy^2) + yz^2) \mathbf{i} \\ & + (2xy \cos(xy^2) + xz^2 + ze^{yz}) \mathbf{j} + (2xyz + ye^{yz} + 2z) \mathbf{k} \end{aligned}$$

is conservative.

- c. Find its potential function.

5. $\mathbf{F}(x, y, z) = (y^2 \cos(xy^2) + yz^2) \mathbf{i} + (2xy \cos(xy^2) + xz^2 + ze^{yz}) \mathbf{j} + (2xyz + ye^{yz} + 2z) \mathbf{k}$

a. Let C be the curve of intersection of the cone $z^2 = 4x^2 + 9y^2$ and the plane $z = 1 + x + 2y$ and D be the part of the curve C that lies in the first octant $x \geq 0, y \geq 0, z \geq 0$ from $(1, 0, 2)$ to $(0, 1, 3)$. Evaluate $\int_D \mathbf{F} \cdot d\mathbf{r}$.

b. Let C be the curve of intersection of $x^2 + y^2 = 1$ and $z = 40$. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$.

6. Evaluate

$$\oint_C \frac{y^2}{x} dx + y \ln x dy$$

where C is the counterclockwise boundary of the region in the first quadrant bounded by the curves $y = x$, $y = \frac{x}{2}$, $y = \frac{1}{x}$ and $\frac{2}{x}$.

1. a. To find the critical points of f , determine where both $f_x = f_y = 0$ or one of the partial derivatives does not exist. Apply the product rule appropriately.

$$f_x = e^y(-2x), \quad f_y = e^y(y^2 - x^2) + e^y(2y) = e^y(y^2 + 2y - x^2)$$

$$f_x = 0 \implies e^y(-2x) = 0, \quad f_y = 0 \implies e^y(y^2 + 2y - x^2) = 0$$

$$e^y \neq 0 \implies x = 0, \quad y^2 + 2y - x^2 = 0 \implies y(y+2) - 0^2 = 0 \implies y = 0, \quad y = -2$$

The critical points occur at $(0, 0)$ and $(0, -2)$. To classify these points, apply the second derivative test.

$$f_{xx} = -2e^y, \quad f_{xy} = f_{yx} = e^y(-2x), \quad f_{yy} = e^y(y^2 + 2y - x^2) + e^y(2y + 2)$$

Calculate the Hessian determinant at these points.

$$\begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = f_{xx}f_{yy} - f_{xy}^2$$

$$(0, 0) \rightarrow \begin{cases} f_{xx} = -2, & f_{xy} = f_{yx} = 0, & f_{yy} = 2 \\ f_{xx}f_{yy} - f_{xy}^2 = -2 \cdot 2 - 0^2 = -4 < 0 \end{cases}$$

$$(0, -2) \rightarrow \begin{cases} f_{xx} = -2e^{-2}, & f_{xy} = f_{yx} = 0, & f_{yy} = -2e^{-2} \\ f_{xx}f_{yy} - f_{xy}^2 = -2e^{-2} \cdot 2e^{-2} - 0^2 = 4e^{-4} > 0, & f_{xx} = -2e^{-2} < 0 \end{cases}$$

A saddle point occurs at $(0, 0)$ and a local maximum occurs at $(0, -2)$.

- b. Let $g(x, y, z) = x + y + z - 1$ be the constraint. Then solve the system of equations below.

$$\left. \begin{array}{l} \nabla f = \lambda \nabla g \\ g(x, y, z) = 0 \end{array} \right\} \quad \begin{array}{l} \nabla f = \langle yz^2, xz^2, 2xyz \rangle = \lambda \langle 1, 1, 1 \rangle = \lambda \nabla g \\ \therefore yz^2 = xz^2 = 2xyz \end{array}$$

$$yz^2 = xz^2 \implies y = x$$

$$yz^2 = 2xyz \implies z = 2x$$

Use the constraint and write y, z in terms of x .

$$x + y + z - 1 = 0 \implies x + x + 2x = 1 \implies 4x = 1 \implies x = \frac{1}{4}$$

$$x = \frac{1}{4} \implies y = \frac{1}{4}, \quad z = \frac{1}{2}$$

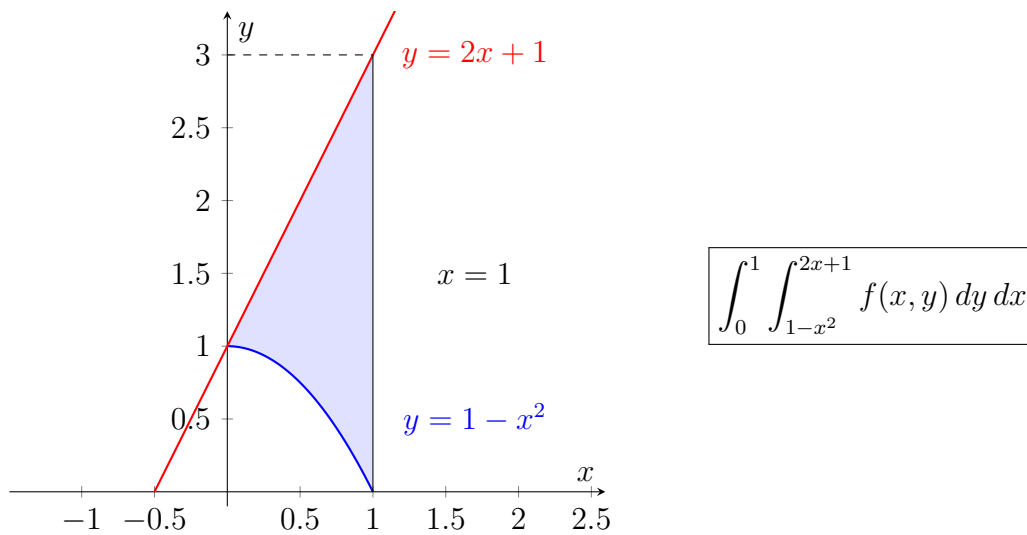
Consider the boundary as well. Set $x = 0, y = 0, z = 0$ one by one. Notice that the value of the function becomes 0 on the boundary of the domain.

Compare all the values.

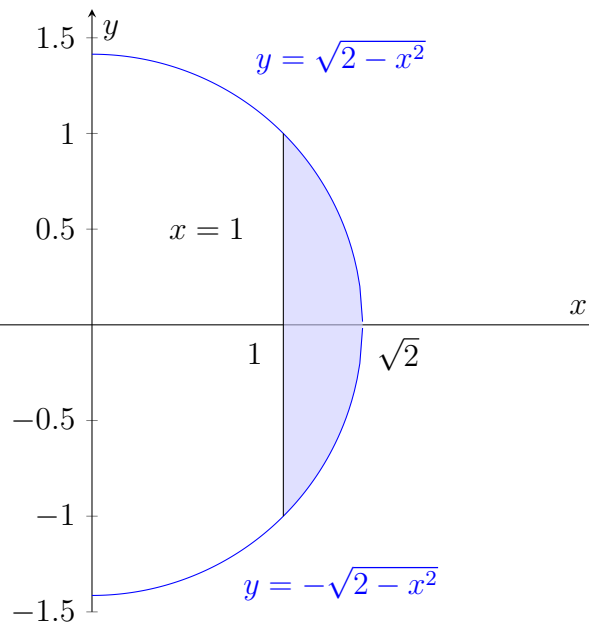
$$f(0, y, z) = f(x, 0, z) = f(x, y, 0) = 0, \quad f\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right) = \frac{1}{4} \cdot \frac{1}{4} \cdot \left(\frac{1}{2}\right)^2 = \frac{1}{64}$$

The maximum value is $\frac{1}{64}$, the minimum value is 0.

2. a.



b. Sketch the region.



$$I = \int_1^{\sqrt{2}} \int_{-\sqrt{2-x^2}}^{\sqrt{2-x^2}} \sqrt{x^2 + y^2} dy dx$$

We may switch to polar coordinates to easily evaluate the integral.

$$\begin{aligned}
 x &= r \cos \theta & x &= \sqrt{2} \cos \theta, \quad y = \sqrt{2} \sin \theta, \quad r_{\text{upper}} = \sqrt{2} \\
 y &= r \sin \theta & x &= 1 \implies r \cos \theta = 1 \implies r_{\text{lower}} = \sec \theta \\
 \theta &= \tan^{-1} \frac{y}{x} & & \\
 x^2 + y^2 &= r^2 & 1 &= \sqrt{2} \cos \theta \implies \cos \theta = \frac{1}{\sqrt{2}} \implies \theta = \pm \frac{\pi}{4} \\
 dA &= dy \, dx = r \, dr \, d\theta & & 0 \leq \theta \leq \pi/2
 \end{aligned}$$

$$\begin{aligned}
 I &= \int_{-\pi/4}^{\pi/4} \int_{\sec \theta}^{\sqrt{2}} r \cdot r \, dr \, d\theta = \int_{-\pi/4}^{\pi/4} \left[\frac{r^3}{3} \right]_{r=\sec \theta}^{r=\sqrt{2}} d\theta = \frac{1}{3} \int_{-\pi/4}^{\pi/4} (2\sqrt{2} - \sec^3 \theta) \, d\theta \\
 &= \int_{-\pi/4}^{\pi/4} \frac{2\sqrt{2}}{3} \, d\theta - \frac{1}{3} \int_{-\pi/4}^{\pi/4} \sec^3 \theta \, d\theta \tag{1}
 \end{aligned}$$

Evaluate the left-hand side integral in (1).

$$\int_{-\pi/4}^{\pi/4} \frac{2\sqrt{2}}{3} \, d\theta = \frac{2\sqrt{2}}{3} \cdot \theta \Big|_{-\pi/4}^{\pi/4} = \frac{\pi\sqrt{2}}{3}$$

Evaluate the right-hand side integral in (1) with the help of integration by parts.

$$\begin{aligned}
 u &= \sec \theta \rightarrow du = \sec \theta \tan \theta \, d\theta \\
 dv &= \sec^2 \theta \, d\theta \rightarrow v = \tan \theta
 \end{aligned}$$

$$\begin{aligned}
 \int \sec^3 \theta \, d\theta &= \tan \theta \cdot \sec \theta - \int \tan^2 \theta \sec \theta \, d\theta = \tan \theta \cdot \sec \theta - \int \frac{1 - \cos^2 \theta}{\cos^3 \theta} \, d\theta \\
 &= \tan \theta \cdot \sec \theta - \int \sec^3 \theta \, d\theta + \int \sec \theta \, d\theta
 \end{aligned}$$

Notice that the integral of $\sec^3 \theta$ with respect to θ appears on the right side of the equation. Perform some algebra and then find the result of the integral.

$$\int \sec^3 \theta \, d\theta = \frac{1}{2} \cdot \tan \theta \cdot \sec \theta + \frac{1}{2} \cdot \int \sec \theta \, d\theta$$

The integral of $\sec \theta$ with respect to θ is as follows.

$$\int \sec \theta \, d\theta = \ln |\tan \theta + \sec \theta| + c_0, \quad c_0 \in \mathbb{R}$$

Rewrite (1).

$$\begin{aligned}
 I &= \frac{\pi\sqrt{2}}{3} - \frac{1}{3} \cdot \frac{1}{2} (\tan \theta \cdot \sec \theta + \ln |\tan \theta + \sec \theta|) \Big|_{-\pi/4}^{\pi/4} \\
 &= \frac{\pi\sqrt{2}}{3} - \frac{1}{6} \left[\left(1 \cdot \sqrt{2} + \ln (1 + \sqrt{2}) \right) - \left(-1 \cdot \sqrt{2} + \ln (-1 + \sqrt{2}) \right) \right] \\
 &= \boxed{\frac{(\pi-1)\sqrt{2}}{3} + \frac{1}{6} \cdot \ln \left(\frac{1+\sqrt{2}}{-1+\sqrt{2}} \right)}
 \end{aligned}$$

3. From the bounds of the integral given in cylindrical coordinates, we infer that this is the region bounded above by the sphere $x^2 + y^2 + z^2 = 4$ and bounded below by the cone $z^2 = x^2 + y^2$

a.

$$\begin{array}{ll}
 z = z & z = \sqrt{4-r^2} \implies z = \sqrt{4-x^2-y^2} \\
 x = r \cos \theta & z = r \implies z = \sqrt{x^2+y^2} \\
 y = r \sin \theta & \rightarrow 0 \leq \theta \leq 2\pi, \quad 0 \leq r \leq \sqrt{2} \\
 r^2 = x^2 + y^2 & \downarrow \\
 dV = r \, dz \, dr \, d\theta = dz \, dy \, dx & -\sqrt{2-y^2} \leq y \leq \sqrt{2-y^2} \\
 & -\sqrt{2} \leq y \leq \sqrt{2}
 \end{array}$$

$$\boxed{\int_{-\sqrt{2}}^{\sqrt{2}} \int_{-\sqrt{2-y^2}}^{\sqrt{2-y^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{4-x^2-y^2}} 3 \, dz \, dx \, dy}$$

- b. For spherical coordinates, we have

$$\begin{array}{ll}
 z = \rho \cos \phi & z = \sqrt{4-r^2} \implies \rho \cos \phi = \sqrt{4-\rho^2 \sin^2 \phi} \\
 r = \rho \sin \phi & \implies \rho_{\text{upper}} = 2 \\
 x^2 + y^2 + z^2 = \rho^2 & \rightarrow z = r \implies \rho \cos \phi = \rho \sin \phi \implies \rho_{\text{lower}} = 0 \\
 dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta & \cos \phi = \sin \phi \implies \tan \phi = 1 \implies \phi_{\text{upper}} = \frac{\pi}{4} \\
 & \phi_{\text{lower}} = 0, \quad 0 \leq \theta \leq 2\pi
 \end{array}$$

$$\boxed{\int_0^{2\pi} \int_0^{\pi/4} \int_0^2 3\rho^2 \sin \phi \, d\rho \, d\phi \, d\theta}$$

4. a. For $\mathbf{F}$ to be conservative, it must be the gradient of some potential function ϕ . We may apply the component test to determine whether mixed partial derivatives are equal.

$$\left. \begin{aligned} \frac{\partial F_1}{\partial y} &= 2x \cos(x^2 y) - 2xy \sin(x^2 y) \cdot x^2 \\ \frac{\partial F_2}{\partial x} &= -2x \cos(x^2 y) + x^2 \sin(x^2 y) \cdot 2xy \end{aligned} \right\} \implies \frac{\partial F_1}{\partial y} \neq \frac{\partial F_2}{\partial x}$$

The mixed partial derivatives are not equal. Therefore, the force is not conservative.

- b. Like what we did above, determine the mixed partial derivatives.

$$\frac{\partial F_1}{\partial y} = 2y \cos(xy^2) - y^2 \sin(xy^2) \cdot 2xy + z^2 = \frac{\partial F_2}{\partial x}$$

$$\frac{\partial F_1}{\partial z} = 2yz = \frac{\partial F_3}{\partial x}$$

$$\frac{\partial F_2}{\partial z} = 2xz + e^{yz} + zye^{yz} = \frac{\partial F_3}{\partial y}$$

- c. Since $\mathbf{F}$ is conservative on $\mathbb{R}^3$, there exists a potential function f such that $\nabla f = \mathbf{F}$.

$$\frac{\partial f}{\partial x} = y^2 \cos(xy^2) + yz^2, \quad \frac{\partial f}{\partial y} = 2xy \cos(xy^2) + xz^2 + ze^{yz}, \quad \frac{\partial f}{\partial z} = 2xyz + ye^{yz} + 2z$$

$$\int \frac{\partial f}{\partial x} dx = \int (y^2 \cos(xy^2) + yz^2) dx = \sin(xy^2) + xyz^2 + g(y, z)$$

Take partial derivative with respect to y .

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} (\sin(xy^2) + xyz^2 + g(y, z)) = 2xy \cos(xy^2) + xz^2 + g_y(y, z)$$

$$= 2xy \cos(xy^2) + xz^2 + ze^{yz} \implies g_y(y, z) = ze^{yz}$$

$$\int \frac{\partial f}{\partial y} dy = \int (2xy \cos(xy^2) + xz^2 + ze^{yz}) dy = \sin(xy^2) + xyz^2 + e^{yz} + h(z)$$

Take partial derivative with respect to z .

$$\frac{\partial f}{\partial z} = \frac{\partial}{\partial z} (\sin(xy^2) + xyz^2 + e^{yz} + h(z)) = 2xyz + ye^{yz} + h_z(z)$$

$$= 2xyz + ye^{yz} + 2z \implies h_z(z) = 2z$$

$$\int \frac{\partial f}{\partial z} dz = \int (2xyz + ye^{yz} + 2z) dz = \sin(xy^2) + xyz^2 + e^{yz} + z^2 + c$$

The potential function for $\mathbf{F}$ is

$$\boxed{f(x, y, z) = \sin(xy^2) + xyz^2 + e^{yz} + z^2 + c, \quad c \in \mathbb{R}}$$

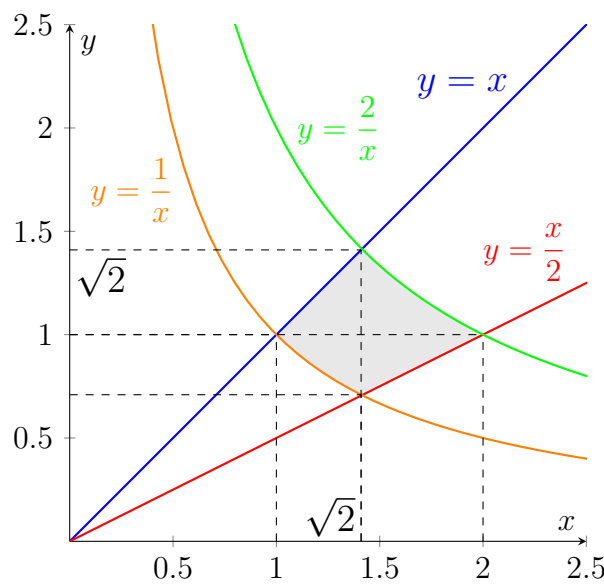
5. a. We showed that $\mathbf{F}$ is conservative in 4(b). Using the Fundamental Theorem of

Line Integrals, evaluate $f(0, 1, 3) - f(1, 0, 2)$.

$$\begin{aligned}
 \int_D \mathbf{F} \cdot d\mathbf{r} &= f(0, 1, 3) - f(1, 0, 2) \\
 &= \sin(0) + 0 + e^3 + 3^2 + c - (\sin(1) + 0 + e^0 + 2^2 + c) \\
 &= \boxed{e^3 + 4}
 \end{aligned}$$

- b. The curve of intersection is a circle, which is a closed curve. Since $\mathbf{F}$ is conservative, the value of the line integral is $\boxed{0}$.

6.



F_1 and F_2 have continuous partial derivatives. C is a closed curve with positive orientation. We may use the tangential form of Green's Theorem to evaluate the line integral.

$$\begin{aligned}
 I &= \oint_C \frac{y^2}{x} dx + y \ln x dy = \iint_R \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dA = \iint_R -\frac{y}{x} dA \\
 &= \int_1^{\sqrt{2}} \int_{1/x}^x -\frac{y}{x} dy dx + \int_{\sqrt{2}}^2 \int_{x/2}^{2/x} -\frac{y}{x} dy dx = \int_1^{\sqrt{2}} -\frac{y^2}{2x} \Big|_{y=1/x}^{y=x} dx + \int_{\sqrt{2}}^2 -\frac{y^2}{2x} \Big|_{y=x/2}^{y=2/x} dx \\
 &= \int_1^{\sqrt{2}} \left(-\frac{x}{2} + \frac{1}{2x^3} \right) dx + \int_{\sqrt{2}}^2 \left(-\frac{2}{x^3} + \frac{x}{8} \right) dx = \left[-\frac{x^2}{4} - \frac{1}{4x^2} \right]_1^{\sqrt{2}} + \left[\frac{1}{x^2} + \frac{x^2}{16} \right]_{\sqrt{2}}^2 \\
 &= \left[-\frac{1}{2} - \frac{1}{8} - \left(-\frac{1}{4} - \frac{1}{4} \right) \right] + \left[\frac{1}{4} + \frac{1}{4} - \left(\frac{1}{2} + \frac{1}{8} \right) \right] = \boxed{-\frac{1}{4}}
 \end{aligned}$$